

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                        |
|----------------------|------------------------|
| United States Patent | 12415533               |
| Kind Code            | B2                     |
| Date of Patent       | September 16, 2025     |
| Inventor(s)          | Kimata; Akihito et al. |

---

### Traffic safety support system

---

#### Abstract

Provided is a traffic safety support system capable of improving traffic safety in a situation in which it is predicted that a mobile body will approach a host vehicle from the front or the rear. A traffic safety support system includes a recognizer, a HMI, a risk notification setting unit, and a risk notification control device. The risk notification setting unit sets a notification mode to a hinting notification mode in a case where a motorcycle is present outside an analogue notification actuation range and acts as an oncoming mobile body or acts as a following mobile body, and the risk notification setting unit sets the notification mode to an analogue notification mode in a case where the motorcycle is present within the analogue notification actuation range.

---

**Inventors:** Kimata; Akihito (Saitama, JP), Inoue; Shigeru (Saitama, JP), Takagi; Yuji (Saitama, JP), Mimura; Yoshitaka (Saitama, JP), Kadowaki; Hideo (Saitama, JP), Kurehashi; Takahiro (Saitama, JP), Okumoto; Masaki (Saitama, JP)

**Applicant:** HONDA MOTOR CO., LTD. (Tokyo, JP)

**Family ID:** 1000008820189

**Assignee:** HONDA MOTOR CO., LTD. (Tokyo, JP)

**Appl. No.:** 18/190042

**Filed:** March 24, 2023

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20230311922 A1   | Oct. 05, 2023    |

#### Foreign Application Priority Data

|    |             |               |
|----|-------------|---------------|
| JP | 2022-061137 | Mar. 31, 2022 |
|----|-------------|---------------|

---

## Publication Classification

**Int. Cl.:** **B60W50/14** (20200101); **B60W10/18** (20120101); **B60W10/20** (20060101);  
**B60W30/095** (20120101); **G08G1/16** (20060101)

**U.S. Cl.:**

**CPC** **B60W50/14** (20130101); **B60W10/18** (20130101); **B60W10/20** (20130101);  
**B60W30/0956** (20130101); **G08G1/166** (20130101); B60W2552/53 (20200201);  
B60W2554/4026 (20200201); B60W2554/4041 (20200201); B60W2554/4044  
(20200201); B60W2555/20 (20200201); B60W2710/18 (20130101); B60W2710/20  
(20130101)

## Field of Classification Search

**CPC:** B60W (50/14); B60W (10/18); B60W (10/20); B60W (30/0956); B60W (2554/4041);  
B60W (2554/4026); B60W (2552/53); B60W (2554/4044); B60W (2555/20); B60W  
(2710/18); B60W (2710/20); G08G (1/166); G01C (21/3476); B62D (1/00); B60T  
(8/1755)

**USPC:** 701/26

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name     | U.S. Cl. | CPC             |
|--------------|-------------|-------------------|----------|-----------------|
| 6720920      | 12/2003     | Breed et al.      | N/A      | N/A             |
| 9434382      | 12/2015     | Prokhorov et al.  | N/A      | N/A             |
| 10460534     | 12/2018     | Brandmaier et al. | N/A      | N/A             |
| 11644834     | 12/2022     | Ditty             | 701/23   | G06F<br>15/7807 |
| 12079006     | 12/2023     | Hill et al.       | N/A      | N/A             |
| 2008/0015772 | 12/2007     | Sanma et al.      | N/A      | N/A             |
| 2014/0368330 | 12/2013     | Watanabe et al.   | N/A      | N/A             |
| 2017/0036673 | 12/2016     | Lee               | N/A      | N/A             |
| 2017/0084177 | 12/2016     | Matsuoka et al.   | N/A      | N/A             |
| 2017/0140651 | 12/2016     | Lee et al.        | N/A      | N/A             |
| 2017/0148318 | 12/2016     | Heckmann          | N/A      | N/A             |
| 2018/0086346 | 12/2017     | Fujisawa et al.   | N/A      | N/A             |
| 2019/0016345 | 12/2018     | Kitagawa          | N/A      | N/A             |
| 2019/0243371 | 12/2018     | Nister et al.     | N/A      | N/A             |
| 2019/0283671 | 12/2018     | Shimomura et al.  | N/A      | N/A             |
| 2019/0283742 | 12/2018     | Kawabe et al.     | N/A      | N/A             |
| 2020/0010082 | 12/2019     | Matsunaga         | N/A      | N/A             |
| 2020/0080863 | 12/2019     | Ichinokawa        | N/A      | N/A             |
| 2022/0165160 | 12/2021     | Maeda et al.      | N/A      | N/A             |
| 2022/0292963 | 12/2021     | Kita              | N/A      | N/A             |
| 2022/0379917 | 12/2021     | Henke et al.      | N/A      | N/A             |

|              |         |                  |       |                 |
|--------------|---------|------------------|-------|-----------------|
| 2022/0383421 | 12/2021 | Grivel et al.    | N/A   | N/A             |
| 2023/0073966 | 12/2022 | Caveney          | N/A   | B60W<br>50/0097 |
| 2023/0076969 | 12/2022 | Kim              | N/A   | G05B 9/02       |
| 2023/0122119 | 12/2022 | Wu et al.        | N/A   | N/A             |
| 2023/0186767 | 12/2022 | Freischem        | 701/2 | B60W 50/06      |
| 2023/0234574 | 12/2022 | Funabashi et al. | N/A   | N/A             |
| 2023/0234575 | 12/2022 | Iwase et al.     | N/A   | N/A             |
| 2023/0311922 | 12/2022 | Kimata et al.    | N/A   | N/A             |
| 2023/0311927 | 12/2022 | Kadowaki et al.  | N/A   | N/A             |
| 2023/0326344 | 12/2022 | Kimata et al.    | N/A   | N/A             |
| 2023/0351895 | 12/2022 | Kadowaki et al.  | N/A   | N/A             |
| 2024/0107376 | 12/2023 | Chakraborty      | N/A   | H04W 4/06       |

**FOREIGN PATENT DOCUMENTS**

| Patent No.    | Application Date | Country | CPC           |
|---------------|------------------|---------|---------------|
| 2944531       | 12/2014          | EP      | B60W<br>30/14 |
| 2009255629    | 12/2008          | JP      | N/A           |
| 2010097261    | 12/2009          | JP      | N/A           |
| 2014146177    | 12/2013          | JP      | N/A           |
| 2017120473    | 12/2016          | JP      | N/A           |
| 2018181061    | 12/2017          | JP      | N/A           |
| 2021136001    | 12/2020          | JP      | N/A           |
| 2021197009    | 12/2020          | JP      | N/A           |
| WO-2020086026 | 12/2019          | WO      | N/A           |

**OTHER PUBLICATIONS**

Notification of Reasons for Refusal issued in the JP Patent Application No. 2022-061148, mailed on Feb. 6, 2024. cited by applicant  
Office Action issued Nov. 6, 2024 in the U.S. Appl. No. 18/190,986. cited by applicant  
Office Action issued Dec. 10, 2024 in the U.S. Appl. No. 18/190,134. cited by applicant  
Office Action issued Mar. 5, 2025 in the U.S. Appl. No. 18/190,077. cited by applicant

*Primary Examiner:* To; Tuan C  
*Attorney, Agent or Firm:* CKC & Partners Co., LLC

**Background/Summary**

(1) This application is based on and claims the benefit of priority from Japanese Patent Application No. 2022-061137, filed on 31 Mar. 2022, the content of which is incorporated herein by reference.  
**BACKGROUND OF THE INVENTION**  
**Field of the Invention**  
(2) The present invention relates to a traffic safety support system. More specifically, the present invention relates to a traffic safety support system that assists a driver in driving a support target which is a mobile body.  
**Related Art**

(3) In public traffic, various traffic participants, such as mobile bodies including four-wheeled automobiles, motorcycles, bicycles, etc. as well as pedestrians, move at different speeds in accordance with their individual intentions. Japanese Unexamined Patent Application, Publication No. 2021-136001 discloses a driving assistance apparatus as a technique for improving safety, convenience, etc. of the traffic participants in public traffic. The driving assistance apparatus performs driving assistance control based on information regarding a traveling state of a vehicle and information regarding a surrounding environment of the vehicle at the time when an occupant of the vehicle has sensed a hazard, in combination with a predicted hazardous state, whereby the driving assistance apparatus gives a warning or executes an intervention in traveling control, without obstructing smooth traveling even when a plurality of objects are present. Patent Document 1: Japanese Unexamined Patent Application, Publication No. 2021-136001

## SUMMARY OF THE INVENTION

(4) According to the conventional driving assistance techniques, a notification is provided to a driver of a host vehicle in a situation in which it is predicted that another mobile body will approach the host vehicle from the front or the rear and there is a possibility that the other mobile body collides with the host vehicle. However, if there is an error in the detection of the position of the other mobile body, an accurate notification cannot be provided to the driver of the host vehicle, and an adequate level of traffic safety cannot be ensured.

(5) It is an object of the present invention to provide a traffic safety support system capable of improving traffic safety in a situation in which it is predicted that a mobile body will approach a host vehicle from the front or the rear. (1) One aspect of the present invention is directed to a traffic safety support system (e.g., a traffic safety support system **1** to be described later) for assisting a driver in driving a support target being a first mobile body (e.g., a four-wheeled automobile **2** to be described later). The traffic safety support system includes: a recognizer (e.g., an on-board driving assistance device **21**, an on-board communication device **24**, a portable information processing terminal **25**, an on-board driving assistance device **31**, an on-board communication device **34**, a portable information processing terminal **35**, a portable information processing terminal **40**, a traffic light control device **55**, an infrastructure camera **56**, a target traffic area recognition unit **60**, and a traffic environment database **67** to be described later) configured to recognize a traffic participant and a traffic environment in a monitoring area surrounding the support target; a human-machine interface (e.g., a HMI **220**, a HMI **320**, and a HMI **420** to be described later) configured to be operated in a manner perceptible to the driver; a notification mode specifier (e.g., a risk notification setting unit **64** to be described later) configured to set a notification mode of the human-machine interface based on a recognition result from the recognizer during a period in which presence of a second mobile body (e.g., a motorcycle **3** to be described later) outside a first range (e.g., an ADAS actuation range to be described later) is recognized by the recognizer, the first range being inside the monitoring area and centered around the support target; and a notification controller (e.g., a risk notification control device **227**, a risk notification control device **327**, and a HMI control device **425** to be described later) configured to operate the human-machine interface in a first notification manner in a case where the notification mode is set to a first mode (e.g., a hinting notification mode to be described later), and operate the human-machine interface in a second notification manner a notification intensity of which is higher than that of the first notification manner in a case where the notification mode is set to a second mode (e.g., an analogue notification mode to be described later). The notification mode specifier sets the notification mode to the first mode in a case where the second mobile body is present outside a second range (e.g., an analogue notification actuation range to be described later) encompassing the first range and acts as an oncoming mobile body approaching the support target from front in a traveling direction of the support target or acts as a following mobile body approaching the support target from rear in the traveling direction of the support target, and the notification mode specifier sets the notification mode to the second mode in a case where the second mobile body is present within the second

range. (2) In the above case, when the notification mode is set to the second mode, the notification mode specifier preferably changes the notification intensity in accordance with a degree of risk between the support target and the second mobile body. (3) In the above case, when at least one of the support target or the oncoming mobile body is partially present in a predetermined width range a center of which is a lane boundary line between the support target and the oncoming mobile body, the notification mode specifier preferably sets the notification mode to the first mode. (4) In the above case, when a nearby mobile body larger than the oncoming mobile body is present near the oncoming mobile body and approaches the support target from the front in the traveling direction of the support target, the notification mode specifier preferably sets the notification mode to the first mode. (5) In the above case, when it is raining in the monitoring area or when an illuminance in the monitoring area is less than a predetermined illuminance threshold, the notification mode specifier preferably sets the notification mode to the first mode. (6) Preferably, the traffic safety support system further includes a predictor (e.g., a prediction unit **62** to be described later) configured to predict futures of mobile bodies recognized as traffic participants by the recognizer. When a first predictive travel route for the support target predicted by the predictor and a second predictive travel route for the second mobile body predicted by the predictor overlap with each other, the notification mode specifier preferably sets the notification mode to the second mode. (7) In the above case, the second mobile body is preferably a two-wheeled vehicle. (8) In the above case, the support target preferably includes a driving assister (e.g., an on-board driving assistance device **21** and an on-board driving assistance device **31** to be described later) configured to automatically operate at least one of a braking device or a steering device of the support target on a condition that a mobile body that has possibility of coming into contact with the support target is present within the first range. (9) Another aspect of the present invention is directed to a traffic safety support system (e.g., a traffic safety support system **1** to be described later) for assisting a driver in driving a support target being a first mobile body (e.g., a four-wheeled automobile **2** to be described later). The traffic safety support system includes: a recognizer (e.g., an on-board driving assistance device **21**, an on-board communication device **24**, a portable information processing terminal **25**, an on-board driving assistance device **31**, an on-board communication device **34**, a portable information processing terminal **35**, a portable information processing terminal **40**, a traffic light control device **55**, an infrastructure camera **56**, a target traffic area recognition unit **60**, and a traffic environment database **67** to be described later) configured to recognize a traffic participant and a traffic environment in a monitoring area surrounding the support target; a human-machine interface (e.g., a HMI **220**, a HMI **320**, and a HMI **420** to be described later) configured to be operated in a manner perceptible to the driver; a notification mode specifier (e.g., a risk notification setting unit **64** to be described later) configured to set a notification mode of the human-machine interface based on a recognition result from the recognizer during a period in which presence of a second mobile body (e.g., a motorcycle **3** to be described later) outside a first range (e.g., an ADAS actuation range to be described later) is recognized by the recognizer, the first range being inside the monitoring area and centered around the support target; and a notification controller (e.g., a risk notification control device **227**, a risk notification control device **327**, and a HMI control device **425** to be described later) configured to operate the human-machine interface in a first notification manner in a case where the notification mode is set to a first mode (e.g., a hinting notification mode to be described later), and operate the human-machine interface in a second notification manner a notification intensity of which is higher than that of the first notification manner in a case where the notification mode is set to a second mode (e.g., an analogue notification mode to be described later). The notification mode specifier sets the notification mode to the first mode in a case where the second mobile body is present outside a second range encompassing the first range and it is difficult for the recognizer to recognize a distance between the support target and the second mobile body, and the notification mode specifier sets the notification mode to the second mode in a case where the second mobile body is present within the

second range. (1) According to the traffic safety support system of the present invention, the notification controller operates the human-machine interface in the first notification manner in a case where the notification mode is set to the first mode, and operates the human-machine interface in the second notification manner a notification intensity of which is higher than that of the first notification manner in a case where the notification mode is set to the second mode, and the notification mode specifier sets the notification mode to the first mode in a case where the second mobile body is present outside the second range and acts as an oncoming mobile body approaching the support target from front in the traveling direction of the support target or acts as a following mobile body approaching the support target from rear in the traveling direction of the support target, whereas the notification mode specifier sets the notification mode to the second mode in a case where the second mobile body is present within the second range. Due to this feature, when it is predicted that the second mobile body will approach the first mobile body from the front or rear of the first mobile body, the traffic safety support system provides a notification in the first notification manner in advance. Further, when the second mobile body has approached the first mobile body, the traffic safety support system provides a notification in the second notification manner. Since the notification intensity of the first notification manner that is provided in advance is less than that of the second notification manner, in the case where it is predicted that the second mobile body will approach the first mobile body from the front or rear of the first mobile body, the traffic safety support system can achieve both mitigation of annoyance caused by the notification to the driver of the first mobile body and securing of the traffic safety. (2) According to the traffic safety support system of the present invention, the notification mode specifier changes the notification intensity in accordance with a degree of risk between the support target and the second mobile body. Due to this feature, when the degree of risk is high, the traffic safety support system can notify with high intensity the approach of the second mobile body to the driver of the support target, whereas when the degree of risk is low, the traffic safety support system can notify the presence of the second mobile body to the driver at a level not annoying the driver. (3) According to the traffic safety support system of the present invention, in a case where at least one of the support target or the oncoming mobile body is present within the predetermined width range, that is, at least one of the support target or the oncoming mobile body is traveling near the lane boundary line and the support target and the oncoming mobile body are substantially frontally approaching each other, the notification mode specifier sets the notification mode to the first mode. Due to this feature, when the support target and the oncoming mobile body are substantially frontally approaching each other, the traffic safety support system according to the present invention provides a notification to the driver in advance, thereby ensuring traffic safety. (4) According to the traffic safety support system of the present invention, the notification mode specifier sets the notification mode to the first mode in a case where a nearby mobile body larger than the oncoming mobile body is present near the oncoming mobile body. Due to this feature, the traffic safety support system of the present invention can provide a notification to the driver in advance when a nearby mobile body larger than the oncoming mobile body is approaching the support target, thereby ensuring the traffic safety. (5) According to the traffic safety support system of the present invention, the notification mode specifier sets the notification mode to the first mode in a case where it is raining in the monitoring area or in a case where an illuminance in the monitoring area is less than a predetermined illuminance threshold. Due to this feature, the traffic safety support system of the present invention can provide a notification in advance when it is raining in the monitoring area or when the illuminance in the monitoring area is low, thereby ensuring the traffic safety. (6) According to the traffic safety support system of the present invention, the notification mode specifier sets the notification mode to the second mode in a case where the first predictive travel route for the support target and the second predictive travel route for the second mobile body overlap with each other, i.e., in a case where a movement vector of the support target and a movement vector of the second mobile body overlap with each other. Due to

this feature, the traffic safety support system of the present invention provides a notification in the second mode in the case where the movement vector of the first mobile body as the support target overlaps with the movement vector of the second mobile body. Thus, the traffic safety support system can provide a notification in the second mode when there is a possibility that the first mobile body and the second mobile body collide with each other, thereby achieving both mitigation of annoyance caused by the notification to the driver and securing of the traffic safety. (7)

According to the traffic safety support system of the present invention, the second mobile body is a two-wheeled vehicle. Due to this feature, in a situation in which the first mobile body as the host vehicle turns right or moves straight ahead while the two-wheeled vehicle as the second mobile body turns right or moves straight ahead, the traffic safety support system can achieve both mitigation of annoyance caused by the notification to the driver and securing of the traffic safety.

(8) According to the traffic safety support system of the present invention, the support target includes the on-board driving assistance device that automatically operates at least one of the braking device or the steering device of the support target on the condition that a mobile body that can come into contact with the support target is present within the first range. Due to this feature, the traffic safety support system of the present invention can reduce, by means of the on-board driving assistance device, the possibility that the first mobile body and the second mobile body collide with each other, and can secure the traffic safety. (9) According to the traffic safety support system of the present invention,

the notification controller operates the human-machine interface in the first notification manner in a case where the notification mode is set to the first mode, and operates the human-machine interface in the second notification manner a notification intensity of which is higher than that of the first notification manner in a case where the notification mode is set to the second mode, and the notification mode specifier sets the notification mode to the first mode in a case where the second mobile body is present outside the second range and it is difficult for the recognizer to recognize a distance between the support target and the second mobile body, whereas the notification mode specifier sets the notification mode to the second mode in a case where the second mobile body is present within the second range. Due to this feature, when it is difficult for the first mobile body as the support target to recognize a distance between the support target and the second mobile body, the traffic safety support system of the present invention can provide a notification in the first notification manner in advance, and can provide a notification in the second notification manner when the support target has approached the second mobile body. Since the notification intensity of the first notification manner in which the notification is provided in advance is less than the notification intensity of the second notification manner, the traffic safety support system of the present invention can achieve both mitigation of annoyance caused by the notification to the driver and securing of the traffic safety.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view illustrating a configuration of a traffic safety support system according to one embodiment of the present invention and part of a target traffic area to be supported by the traffic safety support system;

(2) FIG. 2 is a block diagram illustrating configurations of a coordination support device and a plurality of area terminals communicatively connected to the coordination support device;

(3) FIG. 3A is a block diagram illustrating a configuration of a notification device mountable on a four-wheeled automobile;

(4) FIG. 3B is a block diagram illustrating a configuration of a notification device mountable on a motorcycle;

- (5) FIG. 3C is a block diagram illustrating a configuration of a notification device mountable on a portable information processing terminal held by a pedestrian; and
- (6) FIG. 4 is a schematic diagram for explaining a situation in which a four-wheeled automobile and a motorcycle are in the vicinity of an intersection of roads which is determined as a monitoring area.

## DETAILED DESCRIPTION OF THE INVENTION

- (7) A traffic safety support system according to one embodiment of the present invention will be described below with reference to the drawings.
- (8) FIG. 1 is a view schematically illustrating a configuration of a traffic safety support system **1** according to the present embodiment and part of a target traffic area **9** in which a traffic participant to be supported by the traffic safety support system **1** is present.
- (9) The traffic safety support system **1** assists traffic participants in safely and smoothly moving in the target traffic area **9** in the following manner: the traffic safety support system **1** recognizes, as individual traffic participants, pedestrians **4** that are persons who move in the target traffic area **9** and four-wheeled automobiles **2**, motorcycles **3**, etc. that are mobile bodies which move in the target traffic area **9**, and notifies support information generated through the recognition to each traffic participant to thereby prompt the traffic participants moving according to their intentions to perform mutual communication (specifically, for example, mutual recognition between the traffic participants) and to recognize the surrounding traffic environment.
- (10) FIG. 1 illustrates a case where an area located in an urban district, covering an intersection **52** and the vicinity thereof, and including a road **51**, the intersection **52**, sidewalks **53**, and traffic lights **54** as traffic infrastructure is set as the target traffic area **9**. In the case illustrated in FIG. 1, a total of seven four-wheeled automobiles **2** and a total of two motorcycles **3** move on the road **51** and at the intersection **52**, and a total of three groups of pedestrians **4** move on the sidewalks **53** and at the intersection **52**. Further, in the case illustrated in FIG. 1, a total of three infrastructure cameras **56** are installed.
- (11) The traffic safety support system **1** includes: on-board equipment **20** that moves together with the respective individual four-wheeled automobile **2** (and that includes not only on-board devices mounted on the individual four-wheeled automobile **2** but also a portable information processing terminal held or worn by the driver who drives the individual four-wheeled automobile **2**); on-board equipment **30** that moves together with the respective individual motorcycle **3** (and that includes not only on-board devices mounted on the individual motorcycle **3** but also a portable information processing terminal held or worn by the rider who drives the motorcycle **3**); a portable information processing terminal **40** held or worn by the respective pedestrian **4**; the plurality of infrastructure cameras **56** installed in the target traffic area **9**; a traffic light control device **55** that controls the traffic lights **54**; and a coordination support device **6** communicatively connected to the plurality of terminals present in the target traffic area **9** (hereinafter, also referred to collectively as “area terminals”), including the on-board equipment **20** and **30**, the portable information processing terminals **40**, the infrastructure cameras **56**, the traffic light control device **55**, etc.
- (12) The coordination support device **6** includes one or more computers communicatively connected to the above-described plurality of area terminals via base stations **57**. More specifically, the coordination support device **6** includes a server connected to the plurality of area terminals via the base stations **57**, a network core, and the Internet. Alternatively or additionally, the coordination support device **6** includes, for example, an edge server connected to the plurality of area terminals via the base stations **57** and a multi-access edge computing (MEC) core.
- (13) FIG. 2 is a block diagram illustrating configurations of the coordination support device **6** and the plurality of area terminals communicatively connected to the coordination support device **6**.
- (14) The on-board equipment **20** mounted on the four-wheeled automobile **2** in the target traffic area **9** includes, for example, an on-board driving assistance device **21** that assists a driver in driving the four-wheeled automobile **2**, a notification device **22** that notifies various kinds of



information to the driver, a driving subject state sensor **23** that detects a state of the driver engaged in driving, an on-board communication device **24** that performs wireless communication between the four-wheeled automobile **2** as a host vehicle and the coordination support device **6** and other vehicles near the host vehicle, and a portable information processing terminal **25** held or worn by the driver.

(15) The on-board driving assistance device **21** includes an external sensor unit, a host vehicle state sensor, a navigation device, a driving support ECU, and the like. The external sensor unit includes an exterior camera unit that captures an image of the surroundings of the host vehicle, a plurality of on-board external sensors mounted on the host vehicle, such as a radar and a LIDAR (light detection and ranging) that detect a target outside the vehicle using an electromagnetic wave, and an outside recognition device that acquires information regarding a state around the host vehicle by performing sensor fusion processing on detection results obtained by the foregoing on-board external sensors. The host vehicle state sensor includes sensors that acquire information regarding a traveling state of the host vehicle, such as a vehicle speed sensor, an acceleration sensor, a steering angle sensor, a yaw rate sensor, a position sensor, and an orientation sensor. The navigation device includes, for example, a GNSS receiver that identifies a current position of the host vehicle on the basis of a signal received from a global navigation satellite system (GNSS) satellite, and a storage device that stores map information.

(16) The driving support ECU performs driving support control, such as lane departure prevention control, lane change control, preceding vehicle following control, erroneous start prevention control, collision mitigation brake control, and collision avoidance control, on the basis of the information acquired by the external sensor unit, the host vehicle state sensor, the navigation device, etc. Further, the driving support ECU generates driving assistance information for assisting the driver in safe driving, on the basis of the information acquired by the external sensor unit, the host vehicle state sensor, the navigation device, etc., and transmits the driving assistance information to the notification device **22**.

(17) Here, on the condition that there is a mobile body that may come into contact with the host vehicle within a predetermined collision mitigation brake-actuation range centered around the host vehicle, the driving support ECU starts the collision mitigation brake control to automatically operate a braking device of the host vehicle so as to reduce damage that can be caused by contact of the host vehicle with the mobile body. Further, on the condition that there is a mobile body that may come into contact with the host vehicle within a predetermined collision avoidance steering-actuation range centered around the host vehicle, the driving support ECU starts the collision avoidance control to automatically operate a steering device of the host vehicle so as to avoid contact of the host vehicle with the mobile body. In the following description, the collision mitigation brake-actuation range and the collision avoidance steering-actuation range may be referred to collectively as “ADAS actuation range”.

(18) The driving subject state sensor **23** includes various devices that acquire time-series data of information correlating with driving capability of the driver engaged in driving. For example, the driving subject state sensor **23** includes: an on-board camera that detects a direction of sight line of the driver engaged in driving, whether or not the driver's eyes are open, etc.; a seatbelt sensor that is provided to a seatbelt fastenable on the driver's body, and detects a pulse of the driver, whether or not the driver is breathing, etc.; a steering sensor that is provided to the steering wheel that the driver grips, and detects a skin potential of the driver; and an on-board microphone that detects whether or not there is conversation between the driver and a passenger.

(19) The on-board communication device **24** has a function of transmitting, to the coordination support device **6**, the information acquired by the driving support ECU (including the information acquired by the external sensor unit, the host vehicle state sensor, the navigation device, etc., control information regarding driving support control being executed, and the like), the information regarding the driving subject acquired by the driving subject state sensor **23**, etc. The on-board

communication device **24** further has a function of receiving coordination support information transmitted from the coordination support device **6** and a function of transmitting the received coordination support information to the notification device **22**.

(20) The notification device **22** includes various devices that notify various kinds of information to the driver through, for example, the driver's auditory sense, visual sense, and haptic sense, by causing a human-machine interface (hereinafter, may be abbreviated as "HMI") to operate in a manner that is determined based on the driving assistance information transmitted from the on-board driving assistance device **21** and the coordination support information transmitted from the coordination support device **6**.

(21) FIG. **3A** is a block diagram illustrating a configuration of the notification device **22** mountable on the four-wheeled automobile. Note that FIG. **3A** illustrates, among the components forming the notification device **22**, only blocks involved particularly in control that is performed based on the coordination support information transmitted from the coordination support device **6**.

(22) The notification device **22** includes the HMI **220** that operates in a manner perceptible to the driver, and a HMI control device **225** that causes the HMI **220** to operate based on the coordination support information transmitted from the coordination support device **6**.

(23) The HMI **220** includes an acoustic device **221** that operates in a manner auditorily perceptible to the driver, a head-up display **222** that operates in a manner visually perceptible to the driver, and a seatbelt control device **223** and a seat vibration device **224** that operate in a manner haptically perceptible to the driver.

(24) The acoustic device **221** includes a headrest speaker **221a** that is provided to a headrest of a driver's seat on which the driver sits and is capable of emitting binaural sound having directivity, and a main speaker **221b** that is provided in the vicinity of the driver's seat and a passenger seat. The headrest speaker **221a** and the main speaker **221b** each emit sound according to a command from the HMI control device **225**. The head-up display **222** displays an image according to a command from the HMI control device **225** in a field of view (e.g., a windshield) of the driver engaged in driving. The seatbelt control device **223** changes tension on the seatbelt fastened on the driver's body according to a command from the HMI control device **225**. The seat vibration device **224** vibrates the seat on which the driver sits at an amplitude and/or a frequency according to a command from the HMI control device **225**.

(25) The HMI control device **225** includes a soundness-promoting control device **226** that operates the HMI **220** in a predetermined manner for bringing the driver's driving capability (in particular, cognitive capability) in a sound state, and a risk notification control device **227** that operates the HMI **220** in a predetermined manner for making the driver recognize the presence of an imminent risk. As will be described later, the coordination support information transmitted from the coordination support device **6** to the four-wheeled automobile **2** includes, for example, information regarding a soundness-promoting notification setting value for the soundness-promoting control device **226** to set a soundness-promoting notification to ON or OFF; information regarding a risk notification setting value for the risk notification control device **227** to set a risk notification to ON or OFF and to specify a type of a notification mode (to be described later); and information regarding an imminent risk to the driver (hereinafter, also referred to as "risk information").

(26) The soundness-promoting notification setting value to be inputted to the soundness-promoting control device **226** is set to one of the following values: "0" for the soundness-promoting control device **226** to set the soundness-promoting notification to OFF; and "1" for the soundness-promoting control device **226** to set the soundness-promoting notification to ON.

(27) In a case where the soundness-promoting notification setting value is set to "0", the soundness-promoting control device **226** sets the soundness-promoting notification to OFF. In other words, when the soundness-promoting notification setting value is set to "0", the soundness-promoting control device **226** does not actuate the HMI **220**. Note that this does not prevent the risk notification control device **227** from actuating the HMI **220**.

(28) In a case where the soundness-promoting notification setting value is set to “1”, the soundness-promoting control device **226** sets the soundness-promoting notification to ON. More specifically, for example, the soundness-promoting control device **226** causes the headrest speaker **221a** or the main speaker **221b** to reproduce a piece of music that interests the driver, thereby bringing the driver's driving capability into a sound state. At this time, in order to increase a degree of wakefulness of the driver, the beats per minute (BPM) of the piece of music may be changed or a low-pitched sound may be emphasized.

(29) In this way, the soundness-promoting control device **226** operates the HMI **220** in order to bring the driver's driving capability into a sound state. Therefore, when the risk notification by the risk notification control device **227** to be described later is set to ON (i.e., when the risk notification setting value is set to “1” or “2”), the soundness-promoting notification may be switched to OFF so that the driver will not be annoyed. In the present embodiment, a case will be described in which the soundness-promoting control device **226** brings the driver's driving capability into a sound state mainly through the driver's auditory sense by operating the headrest speaker **221a** and/or the main speaker **221b**, but the present invention is not limited thereto. The soundness-promoting control device **226** may operate, for example, the seatbelt control device **223** and/or the seat vibration device **224**.

(30) The risk notification control device **227** is capable of providing a risk notification in a plurality of notification modes different in terms of the device in the HMI **220** to operate or/and the manner in which the device operates. More specifically, the risk notification control device **227** can provide a risk notification in at least one of the following modes: a hinting notification mode intended to make the driver recognize the presence of a potential risk; an analogue notification mode intended to make the driver recognize the presence of a risk that has surfaced and/or a degree of the risk; and a prediction-assisted notification mode intended to notify to the driver information useful for avoiding a predicted risk. Accordingly, the risk notification setting value to be inputted to the risk notification control device **227** is set to one of the following values: “0” for the risk notification control device **227** to set the risk notification to OFF; “1” for the risk notification control device **227** to set the risk notification to ON and to set the notification mode to the hinting notification mode and the prediction-assisted notification mode; and “2” for the risk notification control device **227** to set the risk notification to ON and to set the notification mode to the analog notification mode and the prediction-assisted notification mode.

(31) In a case where the risk notification setting value is set to “0”, the risk notification control device **227** sets the risk notification to OFF. In other words, when the risk notification setting value is set to “0”, the risk notification control device **227** does not actuate the HMI **220**. Note that this does not prevent the soundness-promoting control device **226** from actuating the HMI **220**.

(32) In a case where the risk notification setting value is set to “1”, the risk notification control device **227** sets the notification mode to the hinting notification mode and the prediction-assisted notification mode, and switches the risk notification to ON in these set notification modes.

(33) In a case where the risk notification setting value is set to “2”, the risk notification control device **227** sets the notification mode to the analogue notification mode and the prediction-assisted notification mode, and switches the risk notification to ON in these set notification modes.

(34) Here, when the notification mode is set to the prediction-assisted notification mode, the risk notification control device **227** generates, based on the risk information transmitted from the coordination support device **6**, risk avoidance assistance information that is useful for the driver to avoid an imminent risk, and operates the acoustic device **221** and/or the head-up display **222** of the HMI **220** in such a manner as to enable the driver to auditorily and/or visually recognize the risk avoidance assistance information. Here, the risk avoidance assistance information includes, for example, information regarding the position of a traffic participant that may come into contact with the host vehicle (hereinafter, referred to also as “risk-carrying subject”), and information with contents that call driver's attention to the risk-carrying subject.

(35) More specifically, in a case where there is a motorcycle driven by a rider in an unsound state in front of the four-wheeled automobile driven by the driver, the risk notification control device **227** causes the acoustic device **221** and/or the head-up display **222** to reproduce and/or display a message such as “Be careful of hazardous right turn of the motorcycle” as the risk avoidance assistance information for enabling avoidance of contact with the motorcycle. At this time, the risk notification control device **227** may cause the head-up display **222** to display an image showing an arrow(s) indicating a current position and/or a predicted position of the motorcycle, as the risk avoidance assistance information for enabling avoidance of contact with the motorcycle.

(36) In a case where the notification mode is set to the hinting notification mode, the risk notification control device **227** operates the HMI **220** in such a manner as not to annoy the driver, thereby making the driver spontaneously recognize the presence of a risk-carrying subject extracted from the risk information transmitted from the coordination support device **6**. Thus, in the hinting notification mode, in order to make the driver spontaneously recognize the presence of the risk-carrying subject without annoying the driver, it is preferable for the risk notification control device **227** to actuate, from among the plurality of devices included in the HMI **220**, the headrest speaker **221a** that appeals particularly to the driver's auditory sense. More specifically, when the notification mode is set to the hinting notification mode, the risk notification control device **227** causes the headrest speaker **221a** to emit, at low volume, a binaural sound effect that is familiar to the driver and is designed to make the driver cast their eyes upon the position of the risk-carrying subject.

(37) In a case where the notification mode is set to the analogue notification, the risk notification control device **227** operates the HMI **220** in a different manner from the manner in the hinting notification mode, thereby making the driver to strongly recognize the presence of a risk-carrying subject extracted from the risk information transmitted from the coordination support device **6** and a degree of risk associated with the risk-carrying subject. Thus, in the analogue notification mode, in order to make the driver strongly recognize the presence of the risk-carrying subject, the risk notification control device **227** operates the HMI **220** in a manner the notification intensity of which is higher than that of the hinting notification mode. Here, the notification intensity refers to intensity for attracting concern and attention of the driver. More specifically, when the notification mode is set to the analogue notification mode, the risk notification control device **227** causes the headrest speaker **221a** and/or the main speaker **221b** to emit buzzer sound or pulse sound at higher volume than the volume of the sound effect emitted in the hinting notification mode. The buzzer sound and the pulse sound are high-volume sound unfamiliar to the driver in comparison with the sound effect emitted in the hinting notification mode, and thus, have higher notification intensity than the sound effect emitted in the hinting notification mode.

(38) When changing the notification intensity according to a degree of risk as described above, the risk notification control device **227** preferably operates the HMI **220** so as to maximize the notification intensity at a point in time at which the above-described driving support ECU starts to execute the collision mitigation brake control or collision avoidance steering control, in other words, at a point in time at which the risk-carrying subject enters the ADAS actuation range of the host vehicle.

(39) Note that while the present embodiment is described with reference to a case in which the risk notification control device **227** actuates the acoustic device **221** when the notification mode is set to the analogue notification mode, the present invention is not limited thereto. When the notification mode is set to the analogue notification mode, the risk notification control device **227** may actuate the seatbelt control device **223** to change tension of the seatbelt and/or actuate the seat vibration device **224** to vibrate the seat, instead of actuating the acoustic device **221**. The seatbelt control device **223** and the seat vibration device **224**, which operate in the manner that appeals to the driver's haptic sense, have higher notification intensity than the sound effect emitted in the hinting notification mode. When the notification mode is set to the analogue notification mode, the risk

notification control device **227** may actuate the acoustic device **221**, the seatbelt control device **223**, and the seat vibration device **224** in combination.

(40) As described above, in the analogue notification mode, in order to make the driver strongly recognize not only the presence of a risk-carrying subject but also a degree of the risk associated with the risk-carrying subject, the risk notification control device **227** preferably changes the notification intensity in accordance with the degree of the risk associated with the risk-carrying subject (e.g., the length of a predicted period until a possible collision with the risk-carrying subject) extracted from the risk information transmitted from the coordination support device **6**. Specifically, the risk notification control device **227** may increase the volume of the buzzer sound, increase the volume of the pulse sound, or shorten the interval of the pulse sounds as the degree of risk increases (i.e., as the predicted period until the possible collision with the risk-carrying subject shortens). When the seatbelt control device **223** is actuated as described above, the risk notification control device **227** may increase the tension on the seatbelt as the degree of risk increases. When the seat vibration device **224** is actuated as described above, the risk notification control device **227** may increase the amplitude of the vibration on the seat as the degree of risk increases.

(41) Referring back to FIG. **2**, the portable information processing terminal **25** includes, for example, a wearable terminal worn by the driver of the four-wheeled automobile **2**, a smartphone held by the driver, and the like. The wearable terminal has a function of measuring biometric information of the driver (e.g., a heart rate, a blood pressure, a blood oxygen level, etc.) and transmitting the measurement data of the biometric information to the coordination support device **6**, and a function of receiving coordination support information transmitted from the coordination support device **6** and notifying to the driver a message according to the coordination support information by means of an image, voice, warning sound, vibration, etc. The smartphone has a function of transmitting information regarding the driver (e.g., position information, a travel acceleration, schedule information, etc.) to the coordination support device **6** and a function of receiving coordination support information transmitted from the coordination support device **6** and notifying to the driver a message according to the coordination support information by means of an image, voice, warning sound, melody, vibration, etc.

(42) The on-board equipment **30** mounted on the motorcycle **3** in the target traffic area **9** includes, for example, an on-board driving assistance device **31** that assists a rider in driving the motorcycle **3**, a notification device **32** that notifies various kinds of information to the rider, a rider state sensor **33** that detects a state of the rider engaged in driving, and a portable information processing terminal **35** held or worn by the rider.

(43) The on-board driving assistance device **31** includes an external sensor unit, a host vehicle state sensor, a navigation device, a driving support ECU, and the like. The external sensor unit includes an exterior camera unit that captures an image of the surroundings of the motorcycle **3** as a host vehicle, a plurality of on-board external sensors mounted on the host vehicle, such as a radar and a LIDAR that detect a target outside the vehicle using an electromagnetic wave, and an outside recognition device that acquires information regarding a state around the host vehicle by performing sensor fusion processing on detection results obtained by the foregoing on-board exterior sensors. The host vehicle state sensor includes sensors that acquire information regarding a traveling state of the host vehicle, such as a vehicle speed sensor and a five-axis or six-axis inertia measurement device. The navigation device includes, for example, a GNSS receiver that identifies a current position on the basis of a signal received from a GNSS satellite, and a storage device that stores map information.

(44) The driving support ECU performs driving support control, such as lane keeping control, lane departure prevention control, lane change control, preceding vehicle following control, erroneous start prevention control, and collision mitigation brake control, on the basis of the information acquired by the external sensor unit, the host vehicle state sensor, the navigation device, etc. Further, the driving support ECU generates driving assistance information for assisting the rider in

safe driving, on the basis of the information acquired by the external sensor unit, the host vehicle state sensor, the navigation device, etc., and transmits the driving assistance information to the notification device **32**.

(45) Here, on the condition that there is a mobile body that may come into contact with the host vehicle within a predetermined collision mitigation brake-actuation range centered around the host vehicle, the driving support ECU starts the collision mitigation brake control to automatically operate a braking device of the host vehicle so as to reduce damage that can be caused by contact of the host vehicle with the mobile body. Further, on the condition that there is a mobile body that may come into contact with the host vehicle within a predetermined collision avoidance steering-actuating range centered around the host vehicle, the driving support ECU starts the collision avoidance control to automatically operate a steering device of the host vehicle so as to avoid contact of the host vehicle with the mobile body. In the following description, the collision mitigation brake-actuation range and the collision avoidance steering-actuation range may be referred to collectively as “ADAS actuation range”.

(46) The rider state sensor **33** includes various devices that acquire information correlating with driving capability of the rider engaged in driving. The rider state sensor **33** includes, for example, a seat sensor that is provided at a seat on which the rider sits and that detects a pulse of the rider, whether or not the rider is breathing, etc.; and a helmet sensor that is provided to a helmet to be worn by the rider and that detects a pulse of the rider, whether or not the rider is breathing, a skin potential, etc.

(47) The on-board communication device **34** has a function of transmitting, to the coordination support device **6**, the information acquired by the driving support ECU (including the information acquired by the external sensor unit, the host vehicle state sensor, the navigation device, etc., control information regarding driving assistance control being executed, and the like), information regarding the rider acquired by the rider state sensor **33**, etc. The on-board communication device **34** further has a function of receiving coordination support information transmitted from the coordination support device **6** and a function of transmitting the received coordination support information to the notification device **32**.

(48) The notification device **32** includes various devices that notify various kinds of information to the rider through, for example, the rider's auditory sense, visual sense, haptic sense, by causing a HMI to operate in a manner determined based on the driving assistance information transmitted from the on-board driving assistance device **31** and the coordination support information transmitted from the coordination support device **6**.

(49) FIG. **3B** is a block diagram illustrating a configuration of the notification device **32** mountable on the motorcycle. Note that FIG. **3B** illustrates, among the components forming the notification device **32**, only blocks involved particularly in control that is performed based on the coordination support information transmitted from the coordination support device **6**.

(50) The notification device **32** includes the HMI **320** that operates in a manner perceptible to the rider, and a HMI control device **325** that causes the HMI **320** to operate based on the coordination support information transmitted from the coordination support device **6**.

(51) The HMI **320** includes a head-mounted speaker **321** that operates in a manner auditorily perceptible to the rider, and a head-up display **322** that operates in a manner visually perceptible to the rider.

(52) The head-mounted speaker **321** is installed in a helmet to be worn by the rider and is capable of emitting binaural sound having directivity. The head-mounted speaker **321** emits sound in accordance with a command from the HMI control device **325**. The head-up display **322** displays an image in accordance with a command from the HMI control device **325** within a field of view (e.g., a shield of the helmet) of the rider engaged in driving.

(53) The HMI control device **325** includes a soundness-promoting control device **326** that operates the HMI **320** in a predetermined manner for bringing the rider's driving capability (in particular,

cognitive capability) in a sound state, and a risk notification control device **327** that operates the HMI **320** in a predetermined manner for making the rider recognize the presence of an imminent risk. As will be described later, the coordination support information transmitted from the coordination support device **6** to the motorcycle **3** includes, for example, information regarding a soundness-promoting notification setting value for the soundness-promoting control device **326** to set a soundness-promoting notification to ON or OFF; information regarding a risk notification setting value for the risk notification control device **327** to set a risk notification to ON or OFF and to specify a type of a notification mode; and information regarding an imminent risk to the rider.

(54) The soundness-promoting notification setting value to be inputted to the soundness-promoting control device **326** is set to one of the following values: “0” for the soundness-promoting control device **326** to set the soundness-promoting notification to OFF; and “1” for the soundness-promoting control device **326** to set the soundness-promoting notification to ON.

(55) In a case where the soundness-promoting notification setting value is set to “0”, the soundness-promoting control device **326** sets the soundness-promoting notification to OFF. In other words, when the soundness-promoting notification setting value is set to “0”, the soundness-promoting control device **326** does not actuate the HMI **320**. Note that this does not prevent the risk notification control device **327** from actuating the HMI **320**.

(56) In a case where the soundness-promoting notification setting value is set to “1”, the soundness-promoting control device **326** sets the soundness-promoting notification to ON. More specifically, for example, the soundness-promoting control device **326** causes the head-mounted speaker **321** to reproduce a piece of music that interests the rider, thereby bringing the rider's driving capability into a sound state. At this time, in order to increase a degree of wakefulness of the rider, the beats per minute (BPM) of the piece of music may be changed or a low-pitched sound may be emphasized.

(57) In this way, the soundness-promoting control device **326** operates the HMI **320** in order to bring the rider's driving capability into a sound state. Therefore, when the risk notification by the risk notification control device **327** to be described later is set to ON (i.e., when the risk notification setting value is set to “1” or “2”), the soundness-promoting notification may be switched to OFF so that the rider will not be annoyed.

(58) The risk notification control device **327** is capable of providing a risk notification in a plurality of notification modes different in terms of the device in the HMI **320** to operate or/and the manner in which the device operates. More specifically, the risk notification control device **327** can provide a risk notification in at least one of the following modes: a hinting notification mode intended to make the rider recognize the presence of a potential risk; an analogue notification mode intended to make the rider recognize the presence of a risk that has surfaced and/or a degree of the risk; and a prediction-assisted notification mode intended to notify to the rider information useful for avoiding a predicted risk. Accordingly, the risk notification setting value to be inputted to the risk notification control device **327** is set to one of the following values: “0” for the risk notification control device **327** to set the risk notification to OFF; “1” for the risk notification control device **327** to set the risk notification to ON and to set the notification mode to the hinting notification mode and the prediction-assisted notification mode; and “2” for the risk notification control device **327** to set the risk notification to ON and to set the notification mode to the analog notification mode and the prediction-assisted notification mode.

(59) In a case where the risk notification setting value is set to “0”, the risk notification control device **327** sets the risk notification to OFF. In other words, when the risk notification setting value is set to “0”, the risk notification control device **327** does not actuate the HMI **320**. Note that this does not prevent the soundness-promoting control device **326** from actuating the HMI **320**.

(60) In a case where the risk notification setting value is set to “1”, the risk notification control device **327** sets the notification mode to the hinting notification mode and the prediction-assisted notification mode, and switches the risk notification to ON in these set notification modes.

(61) In a case where the risk notification setting value is set to “2”, the risk notification control device **327** sets the notification mode to the analogue notification mode and the prediction-assisted notification mode, and switches the risk notification to ON in these set notification modes.

(62) Here, when the notification mode is set to the prediction-assisted notification mode, the risk notification control device **327** generates, based on the risk information transmitted from the coordination support device **6**, risk avoidance assistance information that is useful for the rider to avoid an imminent risk, and causes the head-mounted speaker **321** and/or the head-up display **322** of the HMI **320** to operate in such a manner as to enable the rider to auditorily and/or visually recognize the risk avoidance assistance information. Here, the risk avoidance assistance information includes, for example, information regarding the position of a risk-carrying subject that may come into contact with the motorcycle as the host vehicle, and information with contents that call rider's attention to the risk-carrying subject.

(63) More specifically, in a case where there is a four-wheeled automobile driven by a driver in an unsound state in front of the motorcycle driven by the rider, the risk notification control device **327** causes the head-mounted speaker **321** and/or the head-up display **322** to reproduce and/or display a message such as “Be careful of hazardous right turn of the four-wheeled automobile” as the risk avoidance assistance information for enabling avoidance of contact with the four-wheeled automobile. At this time, the risk notification control device **327** may cause the head-up display **322** to display an image showing an arrow(s) indicating a current position and/or a predicted position of the four-wheeled automobile, as the risk avoidance assistance information for enabling avoidance of contact with the four-wheeled automobile.

(64) In a case where the notification mode is set to the hinting notification mode, the risk notification control device **327** operates the HMI **320** in such a manner as not to annoy the rider, thereby making the rider spontaneously recognize the presence of a risk-carrying subject extracted from the risk information transmitted from the coordination support device **6**. Thus, in the hinting notification mode, in order to make the rider spontaneously recognize the presence of the risk-carrying subject without annoying the rider, it is preferable for the risk notification control device **327** to actuate, from among the plurality of devices included in the HMI **320**, the head-mounted speaker **321** that appeals particularly to the rider's auditory sense. More specifically, when the notification mode is set to the hinting notification mode, the risk notification control device **327** causes the head-mounted speaker **321** to emit, at low volume, a binaural sound effect that is familiar to the rider and is designed to make the rider cast their eyes upon the position of the risk-carrying subject.

(65) In a case where the notification mode is set to the analogue notification mode, the risk notification control device **327** operates the HMI **320** in a different manner from the above-described manner in the hinting notification mode, thereby making the rider to strongly recognize the presence of the risk-carrying subject extracted from the risk information transmitted from the coordination support device **6** and a degree of risk associated with the risk-carrying subject. Thus, in the analogue notification mode, in order to make the rider strongly recognize the presence of the risk-carrying subject, the risk notification control device **327** operates the HMI **320** in a manner the notification intensity of which is higher than that of the hinting notification mode. More specifically, in the case where the notification mode is set to the analogue notification mode, the risk notification control device **327** causes the head-mounted speaker **321** to emit buzzer sound or pulse sound at higher volume than the volume of the sound effect emitted in the hinting notification mode. The buzzer sound and the pulse sound are high-volume sound unfamiliar to the rider in comparison with the sound effect emitted in the hinting notification mode, and thus, have higher notification intensity than the sound effect emitted in the hinting notification mode.

(66) As described above, in the analogue notification mode, in order to make the rider strongly recognize not only the presence of a risk-carrying subject but also a degree of the risk associated with the risk-carrying subject, the risk notification control device **327** preferably changes the



notification intensity in accordance with the degree of the risk associated with the risk-carrying subject (e.g., the length of a predicted period until a possible collision with the risk-carrying subject) extracted from the risk information transmitted from the coordination support device **6**. Specifically, the risk notification control device **327** may increase the volume of the buzzer sound, increase the volume of the pulse sound, or shorten the interval of the pulse sound as the degree of risk increases (i.e., as the predicted period until the possible collision with the risk-carrying subject shortens).

(67) When changing the notification intensity according to a degree of risk as described above, the risk notification control device **327** preferably operates the HMI **320** so as to maximize the notification intensity at a point in time at which the above-described driving support ECU starts to execute the collision mitigation brake control, in other words, at a point in time at which the risk-carrying subject enters the ADAS actuation range of the host vehicle.

(68) Referring back to FIG. **2**, the portable information processing terminal **40** held or worn by the pedestrian **4** in the target traffic area **9** includes, for example, a wearable terminal worn by the pedestrian **4**, a smartphone held by the pedestrian **4**, and the like. The wearable terminal has a function of measuring biometric information of the pedestrian **4** (e.g., a heart rate, a blood pressure, a blood oxygen level, etc.) and transmitting the measurement data of the biometric information to the coordination support device **6**, and a function of receiving the coordination support information transmitted from the coordination support device **6**. The smartphone has a function of transmitting pedestrian information regarding the pedestrian **4** (e.g., position information, a travel acceleration, schedule information, etc. of the pedestrian **4**) to the coordination support device **6** and a function of receiving the coordination support information transmitted from the coordination support device **6**.

(69) The portable information processing terminal **40** includes a notification device **42** that operates a HMI in a predetermined manner based on the received coordination support information to thereby notify various kind of information to the pedestrian through, for example, the pedestrian's auditory sense, visual sense, and haptic sense.

(70) FIG. **3C** is a block diagram illustrating a configuration of the notification device **42** installed in the portable information processing terminal **40**. Note that FIG. **3C** illustrates, among the components forming the notification device **42**, only blocks involved particularly in control that is performed based on the coordination support information transmitted from the coordination support device **6**.

(71) The notification device **42** includes the HMI **420** that operates in a manner perceptible to the pedestrian, and a HMI control device **425** that causes the HMI **420** to operate based on the coordination support information transmitted from the coordination support device **6**.

(72) The HMI **420** includes a speaker **421** that operates in a manner auditorily perceptible to the pedestrian, and a vibrator **424** that operates in a manner haptically perceptible to the pedestrian.

(73) The speaker **421** emits sound in accordance with a command from the HMI control device **425**. The vibrator **424** vibrates the main body of the portable information processing terminal **40** at an amplitude and/or a frequency in accordance with a command from the HMI control device **425**.

(74) As will be described later, the coordination support information transmitted from the coordination support device **6** to the portable information processing terminal **40** held by the pedestrian includes information regarding a risk notification setting value for the HMI control device **425** to set a risk notification to ON or OFF and to specify a type of a notification mode, and information regarding an imminent risk to the pedestrian.

(75) The HMI control device **425** is capable of providing a risk notification in a plurality of notification modes different in terms of the device in the HMI **420** to operate or/and the manner in which the device operates. More specifically, the HMI control device **425** can provide the risk notification in at least one of a hinting notification mode intended to make the pedestrian recognize the presence of a potential risk or an analogue notification mode intended to make the pedestrian

recognize the presence of a risk that has surfaced and/or a degree of the risk. Accordingly, the risk notification setting value to be inputted to the HMI control device **425** is set to one of the following values: “0” for the HMI control device **425** to set the risk notification to OFF; “1” for the HMI control device **425** to set the risk notification to ON and to set the notification mode to the hinting notification mode; and “2” for the HMI control device **425** to set the risk notification to ON and to set the notification mode to the analog notification mode.

(76) In a case where the risk notification setting value is set to “0”, the HMI control device **425** sets the risk notification to OFF. In other words, when the risk notification setting value is set to “0”, the HMI control device **425** does not actuate the HMI **420**.

(77) In a case where the risk notification setting value is set to “1”, the HMI control device **425** sets the notification mode to the hinting notification mode and, and switches the risk notification to ON in the set notification mode.

(78) In a case where the risk notification setting value is set to “2”, the HMI control device **425** sets the notification mode to the analogue notification mode, and switches the risk notification to ON in the set notification mode.

(79) When the notification mode is set to the hinting notification mode, the HMI control device **425** operates the HMI **420** in such a manner as not to annoy the pedestrian, thereby making the pedestrian spontaneously recognize the presence of a risk-carrying subject extracted from the risk information transmitted from the coordination support device **6**. More specifically, when the notification mode is set to the hinting notification mode, the HMI control device **425** actuates the vibrator **424** to thereby vibrate the main body of the portable information processing terminal **40** at predetermined amplitude and frequency.

(80) When the notification mode is set to the analogue notification mode, the HMI control device **425** operates the HMI **420** in a different manner from the above-described manner in the hinting notification mode, thereby making the pedestrian to strongly recognize the presence of the risk-carrying subject extracted from the risk information transmitted from the coordination support device **6** and a degree of risk associated with the risk-carrying subject. Thus, in the analogue notification mode, in order to make the pedestrian strongly recognize the presence of the risk-carrying subject, the HMI control device **425** operates the HMI **420** in a manner the notification intensity of which is higher than that of the hinting notification mode. More specifically, when the notification mode is set to the analogue notification mode, the HMI control device **425** causes the speaker **421** to emit, for example, buzzer sound, pulse sound, and a message informing the pedestrian of the presence of the risk.

(81) As described above, in the analogue notification mode, in order to make the pedestrian strongly recognize not only the presence of a risk-carrying subject but also a degree of the risk associated with the risk-carrying subject, the HMI control device **425** preferably changes the notification intensity in accordance with the degree of the risk associated with the risk-carrying subject (e.g., the length of a predicted period until a possible collision with the risk-carrying subject) extracted from the risk information transmitted from the coordination support device **6**. Specifically, the HMI control device **425** may increase the volume of the buzzer sound, increase the volume of the pulse sound, shorten the interval of the pulse sound, or change the content of the message, as the degree of risk increases (i.e., as the predicted period until the possible collision with the risk-carrying subject shortens).

(82) Referring back to FIG. **2**, the infrastructure cameras **56** capture images of the traffic infrastructure in the target traffic area including the roads, the intersection, and the sidewalks as well as the mobile bodies and the pedestrians that move on the roads, the intersection, the sidewalks, and the like, and transmit the obtained image information to the coordination support device **6**.

(83) The traffic light control device **55** controls the traffic lights and transmits, to the coordination support device **6**, traffic light status information regarding current lighting color of the traffic lights

in the target traffic area, a timing at which the lighting color is switched, and the like.

(84) The coordination support device **6** is a computer that assists the traffic participants in moving safely and smoothly in the target traffic area, by generating, for each of the traffic participants to be assisted, coordination support information for prompting each traffic participant to communicate with other traffic participants and recognize the surrounding traffic environment on the basis of the information acquired from the above-described plurality of area terminals present in the target traffic area, and by notifying the generated coordination support information to the traffic participants. Note that among the plural traffic participants in the target traffic area, the targets to be assisted by the coordination support device **6** according to the present embodiment are traffic participants that have means (e.g., notification devices **22**, **32**, and **42**) for receiving the coordination support information generated by the coordination support device **6** and for causing the respective HMI to operate in a manner specified based on the received coordination support information.

(85) The coordination support device **6** includes: a target traffic area recognition unit **60** for recognizing persons and mobile bodies in the target traffic area as individual traffic participants; a driving subject information acquisition unit **61** for acquiring driving subject state information correlating with the driving capability of the driving subject of each mobile body recognized as the traffic participant by the target traffic area recognition unit **60**; a prediction unit **62** for predicting the futures of the traffic participants in the target traffic area; a soundness-promoting notification setting unit **63** for setting a soundness-promoting notification to ON or OFF for each of the individual traffic participants recognized as the support targets by the target traffic area recognition unit **60**; a risk notification setting unit **64** for setting a notification mode in which a risk notification is provided to each of the individual traffic participants recognized as the support targets by the target traffic area recognition unit **60**; a coordination support information notification unit **65** for transmitting the coordination support information generated for each of the individual traffic participants recognized as the support targets by the target traffic area recognition unit **60**; a traffic environment database **67** where information regarding the traffic environment in the target traffic area is accumulated; and a driving history database **68** where pre-registered information regarding past driving histories of driving subjects are accumulated.

(86) The traffic environment database **67** stores information regarding the traffic environment surrounding the traffic participants in the target traffic area, such as pre-registered map information of the target traffic area (e.g., the width of the road, the number of lanes, the speed limit, the width of the sidewalk, the presence or absence of a guardrail between the road and the sidewalk, the position of a crosswalk, etc.) and risk area information regarding a high risk area with a particularly high risk in the target traffic area. In the following description, the information stored in the traffic environment database **67** may be referred to as registered traffic environment information.

(87) The driving history database **68** stores pre-registered information regarding past driving histories of driving subjects in association with the registration numbers of the mobile bodies possessed by the driving subjects. Therefore, if the registration number of a recognized mobile body is identified by the target traffic area recognizing unit **60** to be described later, the past driving history of the driving subject of the recognized mobile body can be retrieved by way of search by the registration number through the driving history database **68**. In the following description, the information stored in the driving history database **68** may be referred to as registered driving history information.

(88) The target traffic area recognition unit **60** recognizes recognition targets each of which is a traffic participant as a person or a mobile body in the target traffic area, recognizes traffic environment surrounding each traffic participant in the target traffic area, and acquires recognition information regarding the recognition targets, based on the information transmitted from the above-described area terminals (the on-board equipment **20** and **30**, the portable information processing terminals **40**, the infrastructure cameras **56**, and the traffic light control device **55**) in the target

traffic area and the registered traffic environment information read from the traffic environment database **67**.

(89) Here, the information transmitted to the target traffic area recognition unit **60** from the on-board driving assistance device **21** and the on-board communication device **24** included in the on-board equipment **20**, and the information transmitted to the target traffic area recognition unit **60** from the on-board driving assistance device **31** and the on-board communication device **34** included in the on-board equipment **30** include, for example, information regarding traffic participants present near the respective host vehicle and a traffic environmental state that have been acquired by the respective external sensor unit, and information regarding a state of the respective host vehicle as one traffic participant that has been acquired by, for example, the host vehicle state sensor and the navigation device. The information transmitted from the portable information processing terminal **40** to the target traffic area recognition unit **60** includes information regarding a state of the pedestrian as one traffic participant, such as a position and travel acceleration. The image information transmitted from the infrastructure cameras **56** to the target traffic area recognition unit **60** includes information regarding the traffic participants and traffic environment surrounding the traffic participants, such as appearance of the traffic infrastructure such as the roads, the intersection, and the sidewalks, and appearance of traffic participants moving in the target traffic area. The traffic light status information transmitted from the traffic light control device **55** to the target traffic area recognition unit **60** includes information regarding the traffic environment surrounding the traffic participants, such as current lighting color of the traffic lights and a timing for switching the lighting color. The registered traffic environment information that the target traffic area recognition unit **60** reads from the traffic environment database **67** includes information regarding the traffic environment surrounding the traffic participants, such as map information, the risk area information, etc. of the target traffic area.

(90) Thus, the target traffic area recognition unit **60** can acquire, based on the information transmitted from the foregoing area terminals, recognition information regarding each traffic participant (hereinafter, referred to also as “traffic participant recognition information”), such as a position, a moving speed, a moving direction, a moving direction, etc. of each traffic participant in the target traffic area, vehicle type and rank, the registration number, etc. of each mobile body, the number of persons forming a pedestrian group, and age groups, etc. of the pedestrians. The target traffic area recognition unit **60** can further acquire, based on the information transmitted from the foregoing area terminals, recognition information regarding the traffic environment surrounding each traffic participant (hereinafter, referred to also as “traffic environment recognition information”) in the target traffic area, such as the width of the road, the number of lanes, the speed limit, the width of the sidewalk, the presence or absence of a guardrail between the road and the sidewalk, lighting color of the traffic light, a timing for switching the lighting color, and the risk area information.

(91) Accordingly, in the present embodiment, a recognizer that recognizes the traffic participants and the traffic environment in the target traffic area includes: the target traffic area recognition unit **60**; the on-board driving assistance device **21**, the on-board communication device **24**, and the portable information processing terminal **25** that are included in the on-board equipment **20** of the four-wheeled automobile **2**; the on-board driving assistance device **31**, the on-board communication device **34**, and the portable information processing terminal **35** that are included in the on-board equipment **30** of the motorcycle **3**; the portable information processing terminal **40** of the pedestrian **4**; the infrastructure cameras **56**; the traffic light control device **55**; and the traffic environment database **67**.

(92) The target traffic area recognition unit **60** transmits the traffic participant recognition information and the traffic environment recognition information acquired in the manner described above to the driving subject information acquisition unit **61**, the prediction unit **62**, the soundness-promoting notification setting unit **63**, the risk notification setting unit **64**, the coordination support

information notification unit **65**, etc.

(93) The driving subject information acquisition unit **61** acquires driving subject state information and driving subject characteristic information that correlate with current driving capabilities of the driving subjects of the mobile bodies recognized as the traffic participants by the target traffic area recognition unit **60**, based on the information transmitted from the above-described area terminals (in particular, the on-board equipment **20** and **30**) in the target traffic area and the registered driving history information read from the driving history database **68**.

(94) More specifically, in a case where a person is the driving subject of the four-wheeled automobile recognized as the traffic participant by the target traffic area recognition unit **60**, the driving subject information acquisition unit **61** acquires information transmitted from the on-board equipment **20** mounted on the four-wheeled automobile as the driving subject state information of the driver. In a case where a person is the driving subject of the motorcycle recognized as the traffic participant by the target traffic area recognition unit **60**, the driving subject information acquisition unit **61** acquires information transmitted from the on-board equipment **30** mounted on the motorcycle as the driving subject state information of the rider.

(95) Here, the information that is transmitted to the driving subject information acquisition unit **61** from the driving subject state sensor **23** and the on-board communication device **24** included in the on-board equipment **20** contains time-series data correlating with the driving capability of the driver engaged in driving. The time-series data includes, for example, appearance information such as a direction of a sight line of the driver engaged in driving and whether or not the driver's eyes are open, biometric information such as a pulse, whether or not the driver is breathing, and a skin potential, and speech information such as whether or not there is conversation. The information that is transmitted to the driving subject information acquisition unit **61** from the rider state sensor **33** and the on-board communication device **34** included in the on-board equipment **30** contains time-series data correlating with the driving capability of the rider engaged in driving. The time-series data includes, for example, biometric information such as a pulse of the rider, whether or not the rider is breathing, and a skin potential. The information that is transmitted to the driving subject information acquisition unit **61** from the portable information processing terminals **25** and **35** included in the on-board equipment **20** and **30** contains personal schedule information of the driver and that of the rider. For example, in a case where the driver and the rider are driving the respective mobile bodies according to a tight schedule, the driver and the rider may be hasty, and their driving capabilities may degrade. Thus, the personal schedule information of the driver and that of the rider are considered to be information correlating with their driving capabilities.

(96) The driving subject information acquisition unit **61** acquires, by using both or one of the driving subject state information regarding the driving subject acquired through the above procedure and the registered driving history information read from the driving history database **68**, the driving subject characteristic information relating to the driving characteristics of the driving subject (e.g., sudden lane changes with excessive frequency and sudden acceleration and deceleration with excessive frequency) and correlating with the current driving capability of the driving subject engaged in driving.

(97) The driving subject information acquisition unit **61** transmits the driving subject state information and the driving subject characteristic information of the driving subject acquired in the way described above to the prediction unit **62**, the soundness-promoting notification setting unit **63**, the risk notification setting unit **64**, the coordination support information notification unit **65**, etc.

(98) The prediction unit **62** extracts a part of the target traffic area as a monitoring area, and predicts the futures of a plurality of traffic participants in the monitoring area, based on the traffic participant recognition information and the traffic environment recognition information acquired by the target traffic area recognition unit **60** and the driving subject state information and the driving subject characteristic information acquired by the driving subject information acquisition unit **61**. More specifically, the prediction unit **62** constructs a virtual space simulating the monitoring area

based on the traffic participant recognition information and the traffic environment recognition information acquired by the target traffic area recognition unit **60**, and predicts the future of each traffic participant in the monitoring area by performing a simulation in the virtual space based on the traffic participant recognition information, the traffic environment recognition information, the driving subject state information, and the driving subject characteristic information. A detailed description of a specific procedure for the prediction unit **62** to predict the future of each traffic participant in the monitoring area is omitted herein.

(99) Here, the target traffic area has a relatively large range that is determined, for example, in units of municipalities. On the other hand, the monitoring area is a traffic area such as, for example, an area in the vicinity of an intersection or a specific facility, through which a four-wheeled automobile traveling at a legal speed can pass within about a few tens of seconds. In other words, the monitoring area is smaller than the target traffic area, but is larger than the ADAS actuation range set by the driving support ECU mounted on each mobile body.

(100) From among the plural traffic participants present in the target traffic area, the soundness-promoting notification setting unit **63** sets, as setting targets, traffic participants recognized as support targets being mobile bodies by the target traffic area recognition unit **60**, and sets the soundness-promoting notification to ON or OFF for each individual setting target.

(101) More specifically, the soundness-promoting notification setting unit **63** first acquires, from the driving subject information acquisition unit **61**, the driving subject state information and the driving subject characteristic information associated with the driving subject of each setting target that is the mobile body. The soundness-promoting notification setting unit **63** calculates a current degree of soundness of the driving subject for each setting target, based on the acquired driving subject state information and driving subject characteristic information. In a case where the calculated degree of soundness of one setting target is less than a predetermined soundness threshold, the soundness-promoting notification setting unit **63** determines that the driving subject of the one setting target is in an unsound state, and sets the soundness-promoting notification setting value for the setting object to “1” in order to set the soundness-promoting notification for the setting object to ON. In a case where the calculated degree of soundness of one setting target is higher than the predetermined soundness threshold, the soundness-promoting notification setting unit **63** determines that the driving subject of the one setting target is in a sound state, and sets the soundness-promoting notification setting value for the setting object to “0” in order to set the soundness-promoting notification for the setting object to OFF.

(102) The soundness-promoting notification setting unit **63** sets the soundness-promoting notification for each of the plurality of setting targets in the target traffic area to ON or OFF by way of the above-described procedure. The information regarding the soundness-promoting notification setting value that has been set for each setting target by the soundness-promoting notification setting unit **63** is transmitted to the coordination support information notification unit **65**.

(103) From among the plural traffic participants present in the monitoring area that has been extracted from the target traffic area by the prediction unit **62**, the risk notification setting unit **64** determines, as setting targets, the traffic participants recognized as the support targets by the target traffic area recognition unit **60**, and sets the risk notification to ON or OFF and sets a notification mode for each individual setting target.

(104) More specifically, for each individual setting target in the monitoring area, the risk notification setting unit **64** sets the risk notification to ON or OFF and specifies a notification mode based on information related to the monitoring area contained in the traffic participant recognition information and the traffic environment recognition information acquired by the target traffic area recognition unit **60**, information related to the monitoring area contained in the driving subject state information and the driving subject characteristic information acquired by the driving subject information acquisition unit **61**, and prediction results regarding the monitoring area provided by the prediction unit **62**.

(105) In the following description, a specific procedure for the risk notification setting unit **64** to set the risk notification to ON or OFF and to specify a notification mode for each setting target will be described with reference to an example case in which the road **R1** illustrated in FIG. **4** is determined as the monitoring area.

(106) FIG. **4** is a schematic diagram illustrating an intersection of a road **R1** and the vicinity thereof that are determined as a monitoring area, and explains a situation in which a four-wheeled automobile **2** and a motorcycle **3** are present in the monitoring area. In the example illustrated in FIG. **4**, the road **R1** as the monitoring area includes a road having one lane on each side and a road meeting the road at the intersection, and the four-wheeled automobile **2** is traveling on one lane of the road, and the motorcycle **3** is traveling on the other lane opposite to the lane where the four-wheeled automobile **2** is traveling. The four-wheeled automobile **2** is going to move straight ahead through the intersection in the monitoring area, while the motorcycle **3** is going to turn right at the intersection.

(107) In the example illustrated in FIG. **4**, the road **R1** includes the road having one lane on each side and the road meeting the road at the intersection, and the four-wheeled automobile **2** as a support target is traveling on one lane of the road, while the motorcycle **3** as a second mobile body is traveling on the other lane opposite to the lane where the four-wheeled automobile **2** is traveling. The four-wheeled automobile **2** is going to move straight ahead through the intersection in the monitoring area, while the motorcycle **3** is going to turn right at the intersection. In this situation, it is difficult for the driver of the four-wheeled automobile **2** to recognize the distance between the motorcycle **3** and the four-wheeled automobile **2**, and therefore, there is a possibility that the four-wheeled automobile **2** moving straight ahead collides with the motorcycle **3** turning right.

(108) In the example illustrated in FIG. **4**, in a case where the motorcycle **3** is outside the analog notification actuation range encompassing the ADAS actuation range and acts as an oncoming mobile body approaching the four-wheeled automobile **2** from the front in the traveling direction of the four-wheeled automobile **2**, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** of the four-wheeled automobile **2** to the hinting notification mode. When the four-wheeled automobile **2** and the motorcycle **3** have further approached each other and the motorcycle **3** is present within the analog notification actuation range, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the analog notification mode.

(109) The analog notification actuation range encompasses the ADAS actuation range, and is an actuation range where the risk notification is provided in the analog notification mode in response to shortening of the distance between the motorcycle **3** and the four-wheeled automobile **2**. That is, the analog notification actuation range is larger than the ADAS actuation range.

(110) As described above, the risk notification setting unit **64** sets the risk notification setting value to be inputted to each of the risk notification control device **227** and the risk notification control device **327** to “1” to thereby set the notification mode of each of the risk notification control device **227** and the risk notification control device **327** to the hinting notification mode. The risk notification setting unit **64** sets the risk notification setting value to be inputted to each of the risk notification control device **227** and the risk notification control device **327** to “2” to thereby set the notification mode of each of the risk notification control device **227** and the risk notification control device **327** to the analogue notification mode.

(111) In addition to the examples described above, in a case where the motorcycle **3** acts as a following mobile body approaching the four-wheeled automobile **2** as the support target from the rear in the traveling direction of the four-wheeled automobile **2**, the risk notification setting unit **64** may set the notification mode of the risk notification control device **227** to the hinting notification mode. In a case where the motorcycle **3** is present within the analogue notification actuation range, the risk notification setting unit **64** may set the notification mode of the risk notification control device **227** to the analog notification mode.

(112) In a case where at least one of the four-wheeled automobile **2** or the motorcycle **3** is present within a predetermined width range the center of which is a lane boundary line between the four-wheeled automobile **2** and the motorcycle **3**, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the hinting notification mode.

(113) Here, the predetermined width range is a range with reference to which a state is determined in which at least one of the four-wheeled automobile **2** or the motorcycle **3** is traveling near the lane boundary line and the four-wheeled automobile **2** and the motorcycle **3** are substantially frontally approaching each other. In other words, the state in which at least one of the four-wheeled automobile **2** or the motorcycle **3** is present within the predetermined width range means that the four-wheeled automobile **2** and the motorcycle **3** are substantially frontally approaching each other, whereas a state in which neither the four-wheeled automobile **2** nor the motorcycle **3** is present within the predetermined width range means that the four-wheeled automobile **2** and the motorcycle **3** are not frontally approaching each other (i.e., are not offset).

(114) In a case where a nearby mobile body larger than the motorcycle **3** is present near the motorcycle **3** and is approaching the four-wheeled automobile **2** from the front in the traveling direction of the four-wheeled automobile **2**, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the hinting notification mode. Here, the nearby mobile body larger than the motorcycle **3** is a vehicle having a higher vehicle height than the motorcycle **3**, such as a truck, a dump car, a bus, or a tank truck.

(115) In a case where the external sensor unit of the on-board driving assistance device **21** detects that the weather in the monitoring area is inclement, such as rainfall, snowfall, etc., the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the hinting notification mode. In a case where an illuminance in the monitoring area detected by the external sensor unit of the on-board driving assistance device **21** is less than a predetermined illuminance threshold, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the hinting notification mode. Here, the predetermined illuminance threshold is set to, for example, an illuminance that makes it difficult for the driver of the four-wheeled automobile **2** to recognize the surrounding in the field of view.

(116) The prediction unit **62** predicts the futures of the four-wheeled automobile **2** and the motorcycle **3** in the monitoring area based on the traffic participant recognition information and the traffic environment recognition information acquired by the target traffic area recognition unit **60** and the driving subject state information and the driving subject characteristic information acquired by the driving subject information acquisition unit **61**.

(117) More specifically, the prediction unit **62** predicts a first predictive travel route along which the four-wheeled automobile **2** in the monitoring area is predicted to travel in the future and a second predictive travel route along which the motorcycle **3** in the monitoring area is predicted to travel in the future, based on the traffic participant recognition information, the traffic environment recognition information, the driving subject state information, and the driving subject characteristic information.

(118) In a case where the first predictive travel route for the four-wheeled automobile **2** and the second predictive travel route for the motorcycle **3**, which have been predicted by the prediction unit **62**, overlap with each other, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the analog notification mode. For example, in the example illustrated in FIG. **4**, if the first predictive travel route for the four-wheeled automobile **2** going to move straight ahead overlaps with the second predictive travel route for the motorcycle **3** going to turn right, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the analog notification mode.

(119) In the above example, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the hinting notification mode or the analog notification mode, but the present invention is not limited thereto. The risk notification setting unit **64** may set



the risk notification control device **227** and the risk notification control device **327** to the hinting notification mode or the analog notification mode.

(120) Reference is made to FIG. **2** again. Based on the traffic participant recognition information and the traffic environment recognition information acquired by the target traffic area recognition unit **60**, the driving subject state information and the driving subject characteristic information acquired by the driving subject information acquisition unit **61**, the prediction results provided by the prediction unit **62**, the information regarding the soundness-promoting setting value set by the soundness-promoting notification setting unit **63**, and the information regarding the risk setting value set by the risk notification setting unit **64**, the coordination support information notification unit **65** generates coordination support information for prompting the individual traffic participants, which have been recognized as the support targets by the target traffic area recognition unit **60**, to communicate with nearby traffic participants and to recognize the surrounding traffic environment, and then, transmits the generated coordination support information to each traffic participant.

(121) The coordination support information transmitted from the coordination support information notification unit **65** to each support target includes information regarding a soundness-promoting setting value, information regarding a risk notification setting value, and risk information regarding a risk imminent to each support target. Here, the risk information includes, for example, the prediction results provided by the prediction unit **62**, information regarding the position of traffic participants present near each traffic participant.

(122) The traffic safety support system **1** according to the present embodiment exerts the following effects. (1) In the traffic safety support system **1**, the risk notification control device **227** operates the HMI **220** in the first notification manner in a case where the notification mode is set to the hinting notification mode, and operates the HMI **220** in the second notification manner the notification intensity of which is higher than that of the first notification manner in a case where the notification mode is set to the analog notification mode. The risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the hinting notification mode in a case where the motorcycle **3** is present outside the analog notification actuation range and acts as an oncoming mobile body approaching the four-wheeled automobile **2** from the front in the traveling direction of the four-wheeled automobile **2** or acts as a following mobile body approaching the four-wheeled automobile **2** from the rear in the traveling direction of the four-wheeled automobile **2**. The risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the analogue notification mode in a case where the motorcycle **3** is present within the analog notification actuation range. Due to this feature, when it is predicted that the motorcycle **3** will approach the four-wheeled automobile **2** from the front or rear of the four-wheeled automobile **2**, the traffic safety support system **1** can provide a notification in the first notification manner in advance. Further, when the motorcycle **3** has approached the four-wheeled automobile **2**, the traffic safety support system **1** can provide a notification in the second notification manner. Since the notification intensity of the first notification manner that is provided in advance is less than that of the second notification manner, in the case where it is predicted that the motorcycle **3** will approach the four-wheeled automobile **2** from the front or rear of the four-wheeled automobile **2**, the traffic safety support system **1** can achieve both mitigation of annoyance caused by the notification to the driver of the four-wheeled automobile **2** and securing of the traffic safety. (2) In the traffic safety support system **1**, in the case where the notification mode of the risk notification control device **227** is set to the analog notification mode, the risk notification setting unit **64** changes the notification intensity in accordance with a degree of risk between the four-wheeled automobile **2** and the motorcycle **3**. Due to this feature, when the degree of risk is high, the traffic safety support system **1** can notify with high intensity the approach of the motorcycle **3** as the second mobile body to the driver of the four-wheeled automobile **2**, whereas when the degree of risk is low, the traffic safety support system **1** can notify the presence of the second mobile body to the driver at a level not annoying the driver. (3) In the traffic safety support system

1, in a case where at least one of the four-wheeled automobile 2 or the motorcycle 3 is present within the predetermined width range, that is, at least one of the four-wheeled automobile 2 or the motorcycle 3 is traveling near the lane boundary line and the four-wheeled automobile 2 and the motorcycle 3 are substantially frontally approaching each other, the risk notification setting unit 64 sets the notification mode of the risk notification control device 227 to the hinting notification mode. Due to this feature, when the four-wheeled automobile 2 and the motorcycle 3 are substantially frontally approaching each other, the traffic safety support system 1 according to the present embodiment can provide a notification to the driver in advance, thereby ensuring traffic safety. (4) In the traffic safety support system 1, the risk notification setting unit 64 sets the notification mode of the risk notification control device 227 to the hinting notification mode in a case where a nearby mobile body larger than the motorcycle 3 as an oncoming mobile body is present near the motorcycle 3 and is approaching the four-wheeled automobile 2 from the front in the traveling direction of the four-wheeled automobile 2. Due to this feature, the traffic safety support system 1 according to the present embodiment can provide a notification to the driver in advance when a nearby mobile body larger than the motorcycle 3 as the oncoming mobile body is approaching the four-wheeled automobile 2, thereby ensuring the traffic safety. (5) In the traffic safety support system 1, the risk notification setting unit 64 sets the notification mode of the risk notification control device 227 to the hinting notification mode in a case where it is raining in the monitoring area or in a case where an illuminance in the monitoring area is less than a predetermined illuminance threshold. Due to this feature, the traffic safety support system 1 can provide a notification to the driver in advance when it is raining in the monitoring area or when the illuminance in the monitoring area is low, thereby ensuring the traffic safety. (6) The traffic safety support system 1 further includes the prediction unit 62 for predicting the futures of the mobile bodies recognized as the traffic participants by the recognizer, and the risk notification setting unit 64 sets the notification mode of the risk notification control device 227 to the analog notification mode in a case where the first predictive travel route for the four-wheeled automobile 2 and the second predictive travel route for the motorcycle 3, which have been predicted by the prediction unit 62, overlap with each other. The risk notification setting unit 64 sets the notification mode of the risk notification control device 227 to the analogue notification mode when the first predictive travel route for the four-wheeled automobile 2 and the second predictive travel route for the motorcycle 3 overlap with each other, that is, when a movement vector of the four-wheeled automobile 2 and a movement vector of the motorcycle 3 overlap with each other. Due to this feature, the traffic safety support system 1 can provide a notification in the analog notification mode in the case where the movement vector of the four-wheeled automobile 2 as the support target overlaps with the movement vector of the motorcycle 3. That is, the traffic safety support system 1 can provide a notification in the analog notification mode when there is a possibility that the four-wheeled automobile 2 and the motorcycle 3 collide with each other, thereby achieving both mitigation of annoyance caused by the notification to the driver and securing of the traffic safety. (7) In the traffic safety support system 1, the second mobile body is a two-wheeled vehicle such as a motorcycle 3. Due to this feature, in a situation in which the host vehicle being the first mobile body turns right or moves straight ahead while the motorcycle 3 being the second mobile body turns right or moves straight ahead, the traffic safety support system 1 can achieve both mitigation of annoyance caused by the notification to the driver and securing of the traffic safety. (8) In the traffic safety support system 1, the four-wheeled automobile 2, which is a support target, includes the on-board driving assistance device 21 that automatically operates at least one of a braking device or a steering device of the four-wheeled automobile 2 on the condition that a mobile body that may come into contact with the four-wheeled automobile 2 is present within the ADAS actuation range. Due to this feature, the traffic safety support system 1 can reduce, by means of the on-board driving assistance device 21, the possibility that the four-wheeled automobile 2 and the motorcycle 3 collide with each other, and can secure the traffic safety. (9) In the traffic safety

support system **1**, the risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the hinting notification mode in a case where the motorcycle **3** is present outside the analog notification actuation range encompassing the ADAS actuation range and it is difficult for the recognizer to recognize the distance between the motorcycle **3** and the four-wheeled automobile **2**. The risk notification setting unit **64** sets the notification mode of the risk notification control device **227** to the analogue notification mode in a case the motorcycle **3** is present in the analog notification actuation range. Due to this feature, when it is difficult for the four-wheeled automobile **2** to recognize the distance between the four-wheeled automobile **2** and the motorcycle **3**, the traffic safety support system **1** can provide a notification in the first notification manner in advance, and can provide a notification in the second notification manner when the four-wheeled automobile **2** has approached the motorcycle **3**. Since the notification intensity of the first notification manner in which the notification is provided in advance is less than the notification intensity of the second notification manner, the traffic safety support system **1** can achieve both mitigation of annoyance caused by the notification to the driver and securing of the traffic safety.

(123) While one embodiment of the present invention had been described above, it should be noted that the present invention is not limited to the embodiment described above. Appropriate modification may be made to the specifics of the present invention without deviating from the spirit of the present invention. For example, the above embodiment has been described with reference to the case where the coordination support device **6** capable of wireless communication with the support target includes a recognizer for recognizing the traffic participants and a traffic environment in the monitoring area surrounding the support target that is a mobile body and the notification mode specifier for setting the notification mode for the support target, as the target traffic area recognition unit **60** and the risk notification setting unit **64**, respectively. However, the present invention is not limited to this case. The recognizer and the notification mode specifier may be constituted by on-board devices mounted on the support target. In this case, the range of the monitoring area recognized by the recognizer is limited to a range recognizable by the external sensor mounted on the support target, but there is an advantage that delay due to communication is small.

#### EXPLANATION OF REFERENCE NUMERALS

(124) **1**: Traffic safety support system **9**: Target traffic area **2**: Four-wheeled automobile (First mobile body, Traffic participant) **20**: On-board equipment **21**: On-board driving assistance device (Recognizer, Driving assistor) **22**: Notification device **23**: Driving subject state sensor **24**: On-board communication device (Recognizer) **25**: Portable information processing terminal (Recognizer) **3**: Motorcycle (Second mobile body, Traffic participant) **30**: On-board equipment **31**: On-board driving assistance device (Recognizer, Driving assistor) **32**: Notification device **33**: Rider state sensor **34**: On-board communication device (Recognizer) **35**: Portable information processing terminal (Recognizer) **4**: Pedestrian (Person, Traffic participant) **40**: Portable information processing terminal (Recognizer) **51**: Road (Traffic environment) **52**: Intersection (Traffic environment) **53**: Sidewalk (Traffic environment) **54**: Traffic light (Traffic environment) **55**: Traffic light control device (Recognizer) **56**: Infrastructure camera (Recognizer) **6**: Coordination support device **60**: Target traffic area recognition unit (Recognizer) **61**: Driving subject information acquisition unit **62**: Prediction unit (Predictor) **63**: Soundness-promoting notification setting unit **64**: Risk notification setting unit (Notification mode specifier) **65**: Coordination support information notification unit **67**: Traffic environment database (Recognizer) **68**: Driving history database **220**: HMI (Human-machine interface) **225**: HMI control device **226**: Soundness-promoting control device **227**: Risk notification control device (Notification controller) **320**: HMI (Human-machine interface) **325**: HMI control device **326**: Soundness-promoting control device **327**: Risk notification control device (Notification controller) **420**: HMI (Human-machine interface) **425**: HMI control device

## Claims

1. A traffic safety support system for assisting a driver in driving a support target being a first mobile body, the traffic safety support system comprising: a processor; wherein the processor is configured to: recognize a traffic participant and a traffic environment in a monitoring area surrounding the support target; operate a human-machine interface in a manner perceptible to the driver; set a notification mode of the human-machine interface based on a recognition result during a period in which presence of a second mobile body outside a first range is recognized, the first range being inside the monitoring area and centered around the support target; and operate the human-machine interface in a first notification manner in a case where the notification mode is set to a first mode, and operate the human-machine interface in a second notification manner a notification intensity of which is higher than that of the first notification manner in a case where the notification mode is set to a second mode, wherein the processor sets the notification mode to the first mode in a case where the second mobile body is present outside a second range encompassing the first range and acts as an oncoming mobile body approaching the support target from front in a traveling direction of the support target or acts as a following mobile body approaching the support target from rear in the traveling direction of the support target, and the processor sets the notification mode to the second mode in a case where the second mobile body is present within the second range.
2. The traffic safety support system according to claim 1, wherein in the case where the notification mode is set to the second mode, the processor changes the notification intensity in accordance with a degree of risk between the support target and the second mobile body.
3. The traffic safety support system according to claim 1, wherein in a case where at least one of the support target or the oncoming mobile body is partially present in a predetermined width range a center of which is a lane boundary line between the support target and the oncoming mobile body, the processor sets the notification mode to the first mode.
4. The traffic safety support system according to claim 1, wherein in a case where a nearby mobile body larger than the oncoming mobile body is present near the oncoming mobile body and approaches the support target from the front in the traveling direction of the support target, the processor sets the notification mode to the first mode.
5. The traffic safety support system according to claim 1, wherein in a case where it is raining in the monitoring area or in a case where an illuminance in the monitoring area is less than a predetermined illuminance threshold, the processor sets the notification mode to the first mode.
6. The traffic safety support system according to claim 1, the processor predicts futures of mobile bodies recognized as traffic participants, wherein in a case where a first predictive travel route for the support target and a second predictive travel route for the second mobile body overlap with each other, the processor sets the notification mode to the second mode.
7. The traffic safety support system according to claim 1, wherein the second mobile body is a two-wheeled vehicle.
8. The traffic safety support system according to claim 1, wherein the support target comprises is configured to automatically operate at least one of a braking device or a steering device of the support target on a condition that a mobile body that has possibility of coming into contact with the support target is present within the first range.
9. A traffic safety support system for assisting a driver in driving a support target being a first mobile body, the traffic safety support system comprising: a processor; wherein the processor is configured to: recognize a traffic participant and a traffic environment in a monitoring area surrounding the support target; operate a human-machine interface in a manner perceptible to the driver; set a notification mode of the human-machine interface based on a recognition result during a period in which presence of a second mobile body outside a first range is recognized, the first

range being inside the monitoring area and centered around the support target; and operate the human-machine interface in a first notification manner in a case where the notification mode is set to a first mode, and operate the human-machine interface in a second notification manner a notification intensity of which is higher than that of the first notification manner in a case where the notification mode is set to a second mode, wherein the processor sets the notification mode to the first mode in a case where the second mobile body is present outside a second range encompassing the first range and it is difficult to recognize a distance between the support target and the second mobile body, and the processor sets the notification mode to the second mode in a case where the second mobile body is present within the second range.

---